

Making Quantum Dots and Nanotubes

KKKC6353 Nanoelectronic Devices · Lecture 2, Part 3 of 3

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1

Two routes to the nanoscale

Carve it like a sculptor, or let it grow like a crystal.

Top-down and bottom-up

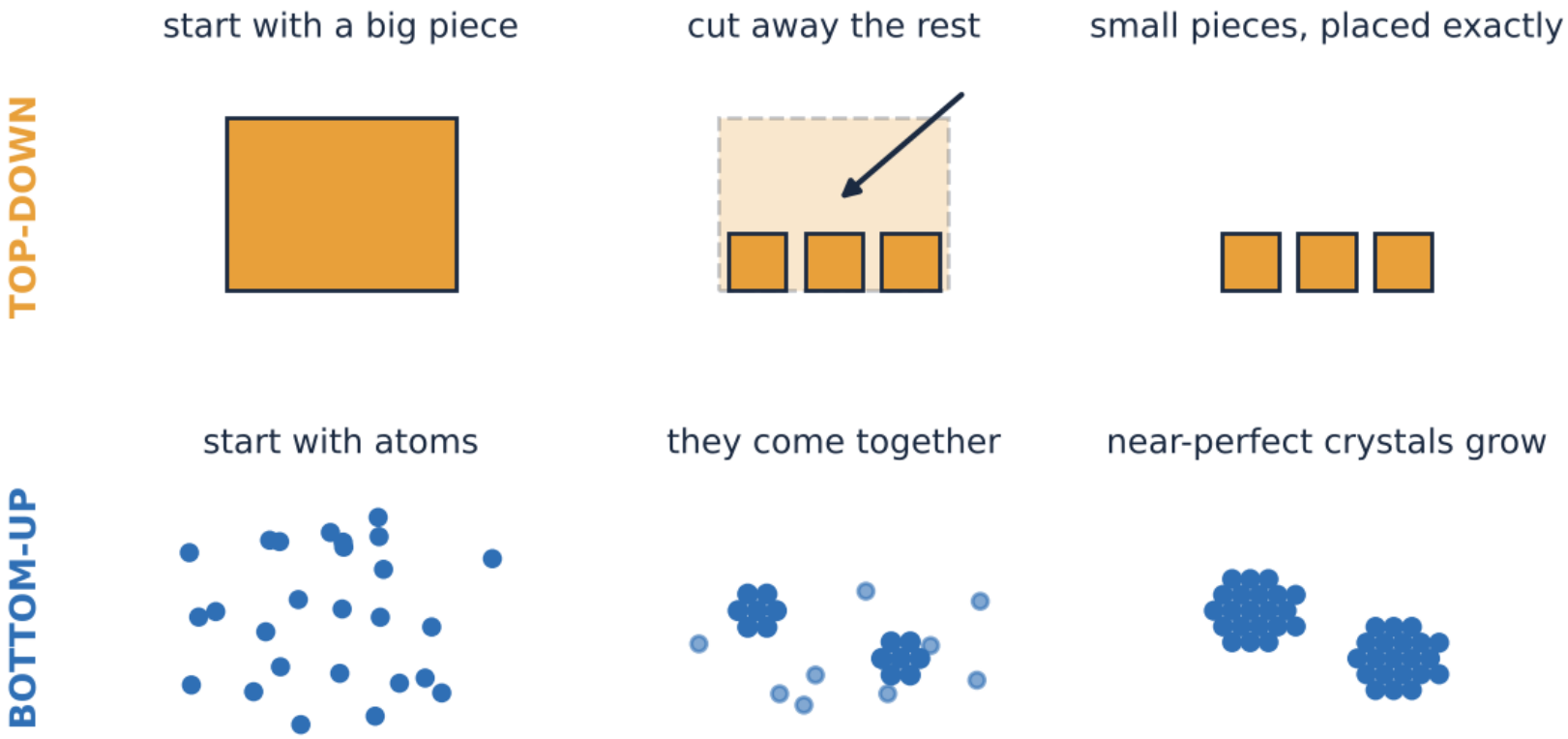


Figure made for this course.

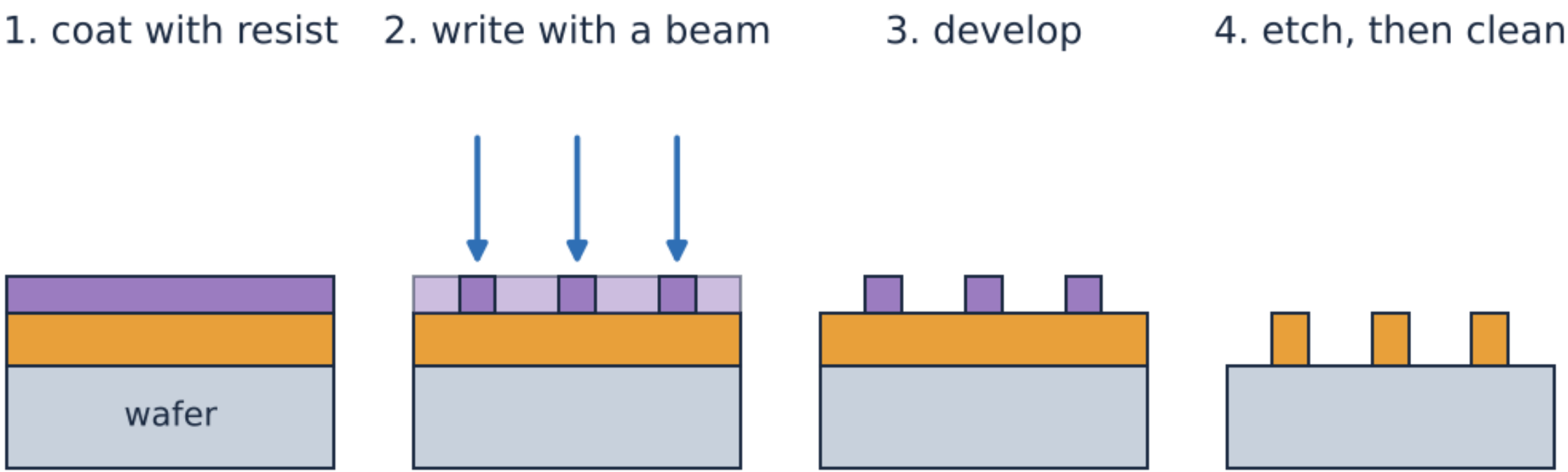
Top-down puts structures exactly where they were drawn. **Bottom-up** makes near-perfect crystals, billions at once. The best devices often combine the two.

2

Top-down: carving

Draw a pattern, then cut it into the material.

Lithography and etching in four steps



Drawn schematically. Figure made for this course.

A **resist** coating records the pattern; after developing, it protects the material underneath while the rest is **etched** away.

Drawing the pattern

Electron-beam lithography

A finely focused electron beam writes like a very fine pen: features below 10 nm, but one point at a time. Research and one-off devices.

Photolithography

Light through a stencil (mask) prints the whole pattern at once: how chips are made in billions. Today's machines use 13.5 nm extreme-ultraviolet light.

Cutting, and its catch

Etching

Dry (plasma) etching cuts steep walls; wet (chemical) etching is gentler but less precise.

The catch: damaged edges

Carved walls are rough at the atomic scale. In a 20 nm dot most atoms are near them, so etched dots glow poorly.

Other tools

Focused ion beams blast material away directly. Dots made by voltage avoid cutting altogether: quantum bits (Lecture 3).

3

Bottom-up I: growing crystals layer by layer

Quantum wells are grown, not carved.

Building with atoms: epitaxy

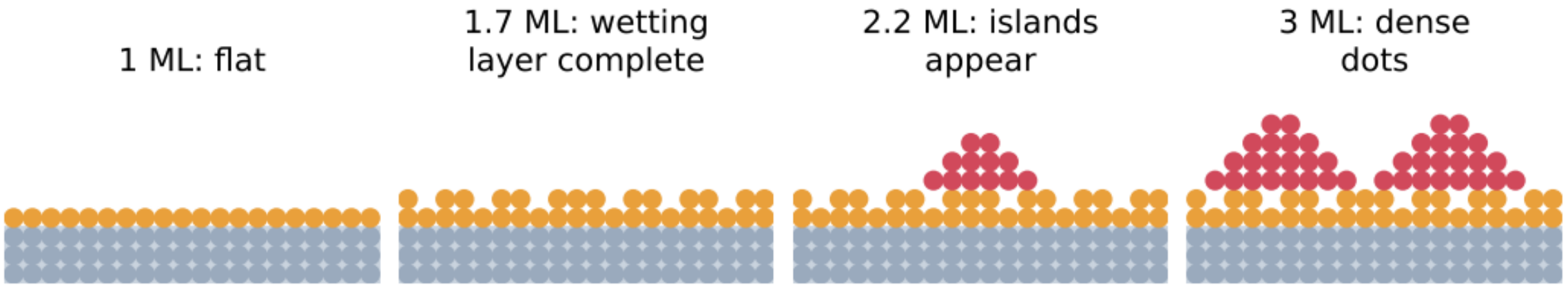
Molecular-beam epitaxy

MBE: beams of atoms in an extremely high vacuum build the crystal about one layer of atoms per second. Shutters give boundaries sharp to a single layer.

Metal-organic CVD (MOCVD)

Gases carrying the atoms react on hot wafers. Faster, many wafers at once: almost every LED and laser diode is grown this way.

Dots that assemble themselves



Drawn schematically. Figure made for this course.

InAs on GaAs (7% mismatch) is **squeezed** to fit, like a carpet too big for its room. After about **1.7 layers** of atoms the film bunches up into islands: **self-assembled dots**.

A common misconception

Not a hybrid

Stranski–Krastanov growth is purely bottom-up: no pattern is drawn and nothing is removed. The dots assemble themselves.

Not ten layers

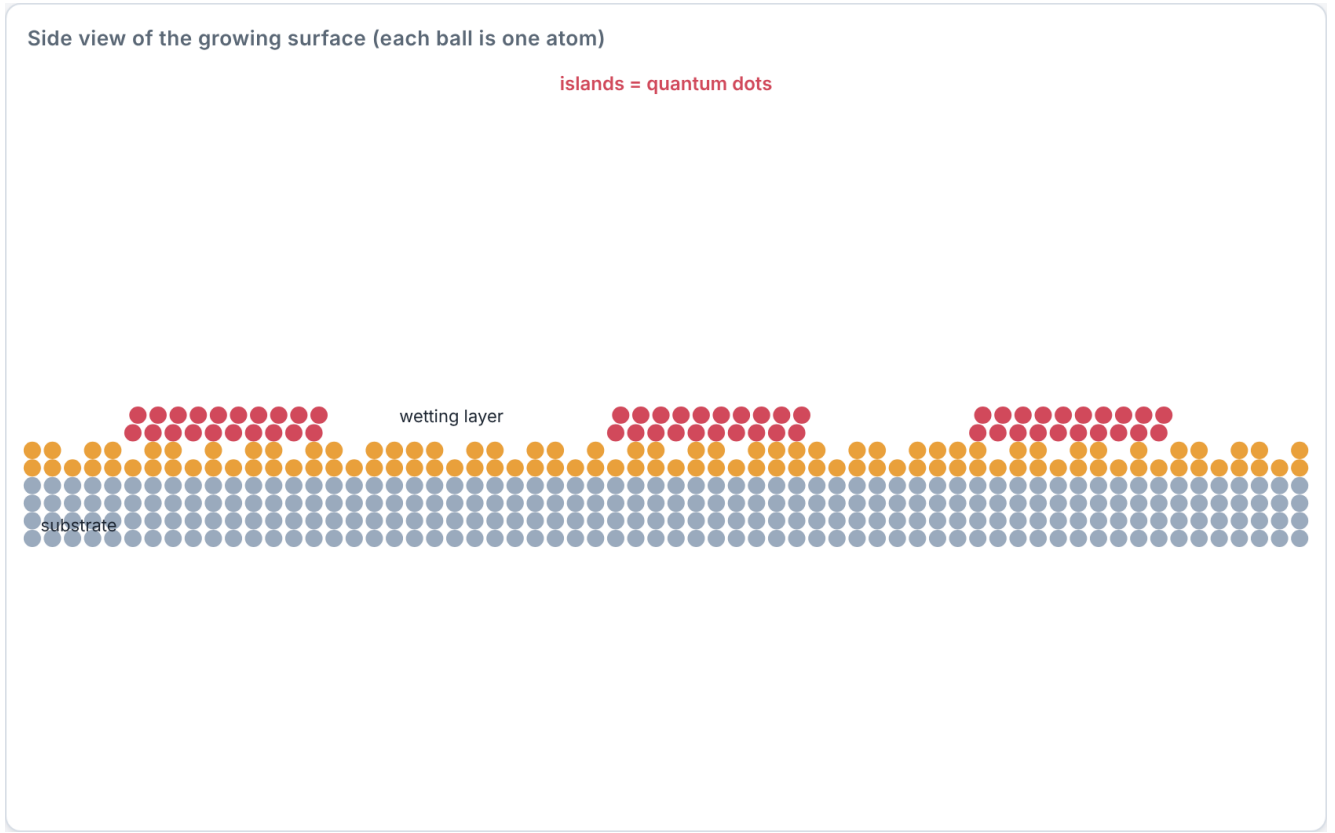
For InAs on GaAs the islands appear after less than two layers of atoms (about 1.7), not after ten.

Interactive animation 7

Growing quantum dots

faizverse.github.io/nanoelectronic-devices/L2/quantum-dot-growth.html

- On a crystal: add material slowly. When do islands appear?
- Does the wetting layer thicken after that?
- In a hot flask: press Inject and watch the colour change
- Quench at green, then try again and stop at red

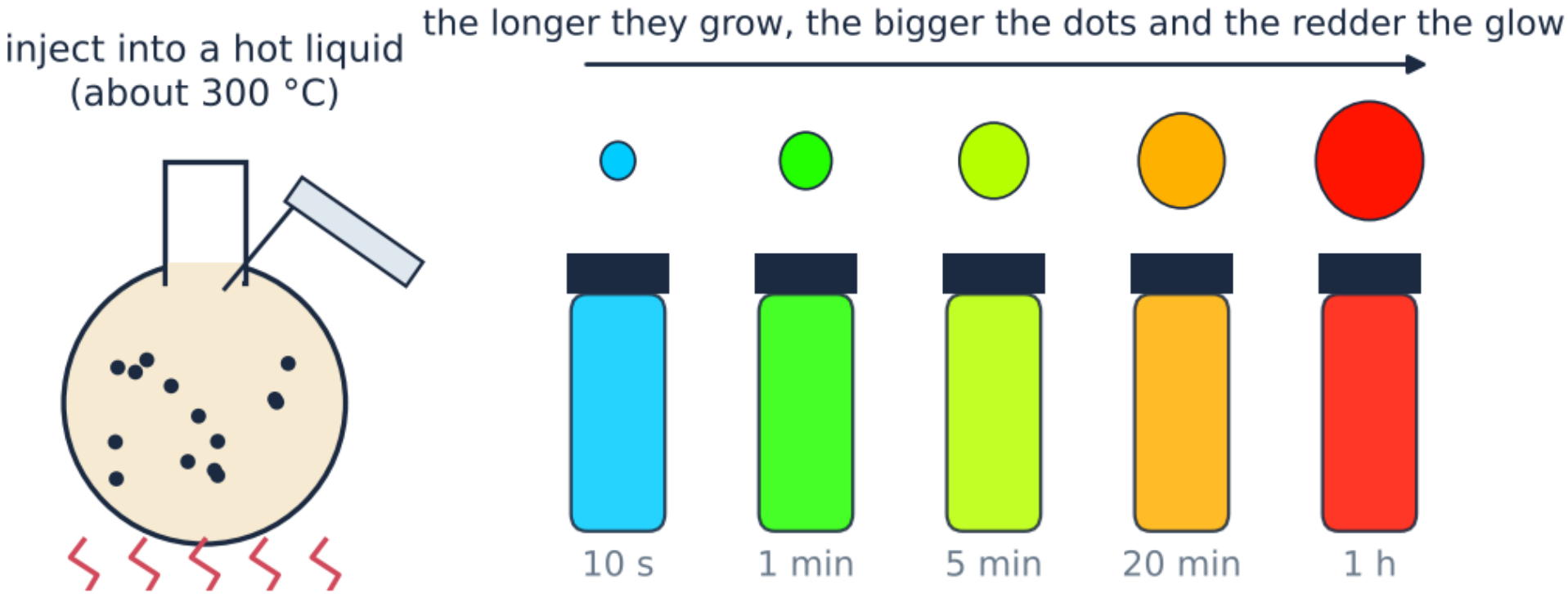


4

Bottom-up II: chemistry in a flask

Most quantum dots in products are made in a hot liquid.

Hot injection



Times are illustrative. Figure made for this course.

All the seeds are born in one **burst**, then grow together, so they stay nearly the same size (often within 5%). **Longer growth → bigger dots → redder glow.**

The recipe

1 Prepare

Two solutions: one with the metal, one with the other element.

2 Inject

One into the other, quickly, at about 300 °C.

3 Seed

A burst of tiny seed crystals in a fraction of a second.

4 Grow

Lower the temperature a little: seeds grow, no new ones form.

5 Stop

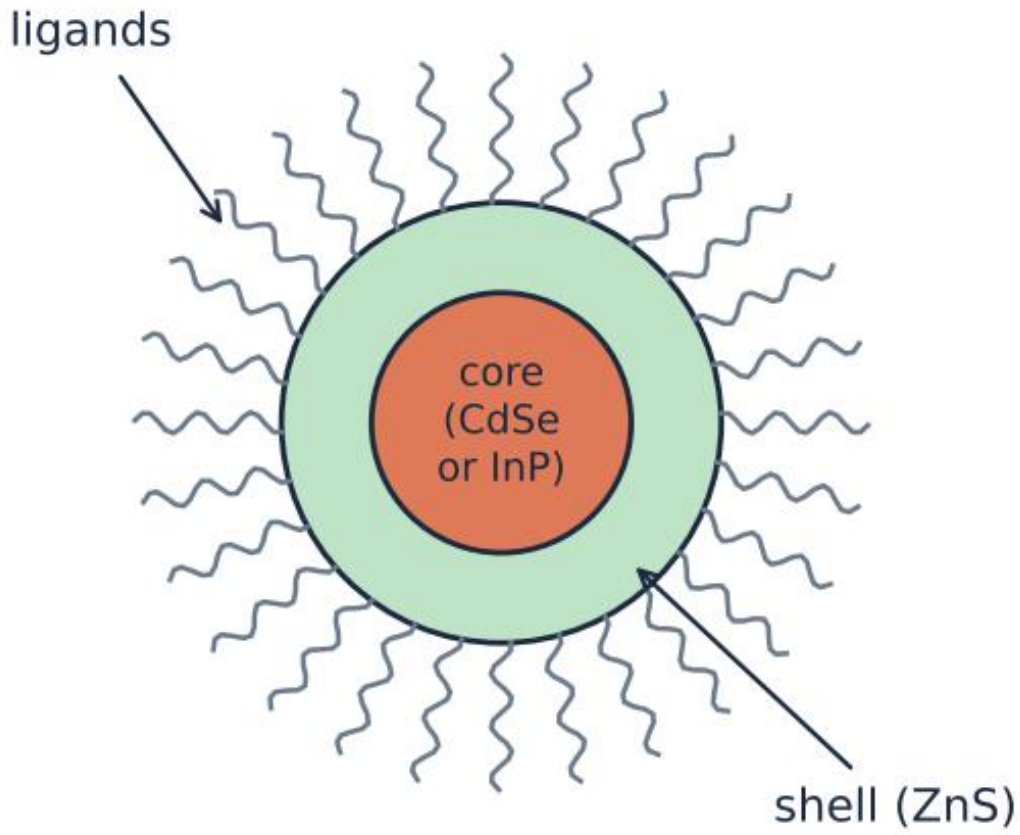
Cool the flask when the dots reach the colour you want.

6 Clean

Settle the dots out of the liquid and wash them.

A coat and some hair

- Bare surfaces trap electrons and stop the glow
- A **shell** (such as ZnS) keeps the electron and hole in the **core**
- Light given out rises from a few per cent to **over 90%**
- **Ligands**, like hairs, keep dots apart and let them be mixed into inks



Drawn schematically. Figure made for this course.

Other recipes (from the L2 handout)

Method	How it works	Typical use
Heat-up	Mix cold, then heat: simpler to scale up	Display dots (CdSe, InP)
Reprecipitation	Dissolve in one liquid, drop into another	Perovskite dots
Hydrothermal	Heat in a sealed, high-pressure vessel	Oxide, sulfide, carbon dots
Microemulsion	Tiny water droplets in oil act as reactors	ZnS, CdS, magnetic dots
Plasma	Dots condense from a glowing gas	Silicon, germanium dots
Flow reactors	Mix and heat in narrow channels, continuously	Large-scale production
Green synthesis	Plants, microbes or biomolecules	Carbon dots

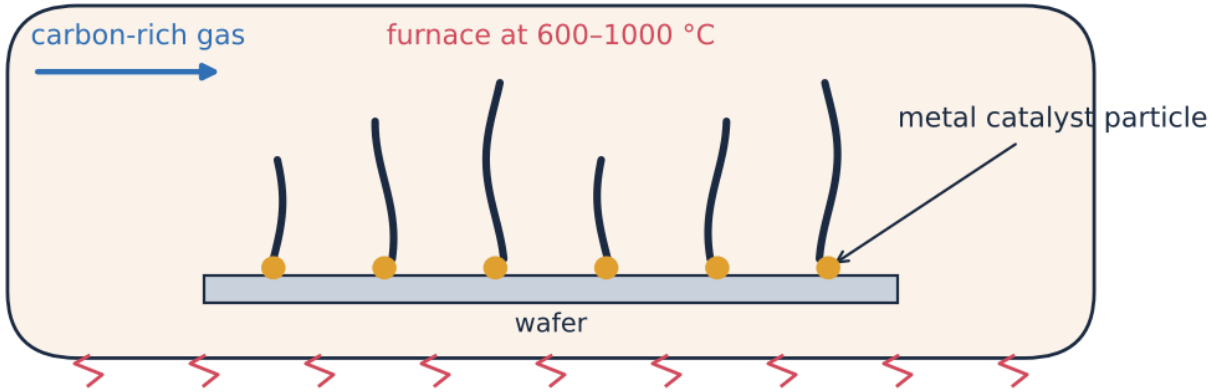
5

Making carbon nanotubes

Grow them, then sort them.

Three ways to grow them

- **Arc discharge**: a big spark between carbon rods (Iijima, 1991)
- **Laser ablation**: a laser vaporises carbon with a little metal (Rice, mid-1990s)
- **Chemical vapour deposition**: carbon-rich gas over metal particles at 600–1000 °C
- CVD is most used today: it scales up and grows tubes on wafers



Drawn schematically. Figure made for this course.

Sorting out the metallic tubes

About one tube in three is metallic, and one metallic tube can short a transistor.

1

Spinning

In a fast centrifuge, coated tubes settle at different depths by type and diameter.

2

Wrapping

Certain polymers wrap only semiconducting tubes and carry them into solution.

3

Burning off

A large current burns away metallic tubes while the semiconducting ones are switched off.

6

Choosing a method

No method is best at everything.

Comparing the methods

Method	Size control	Crystal quality	Placement	Volume	Main uses
E-beam lithography + etching	Good	Poor at edges	Exact	Very low	Research devices
Dots made by voltage	Set by voltages	Excellent	Exact	Low	Quantum bits
MBE / MOCVD wells	One atomic layer	Excellent	Whole wafer	High	Lasers, LEDs, HEMTs
Self-assembled dots	Moderate	Excellent	Random	High	Lasers, single photons
Colloidal (hot injection)	Excellent	High, with shell	None	Very high	Displays, imaging
Nanotube CVD + sorting	Set by catalyst	High	Improving	Medium	Nanotube transistors

The application chooses

A television

Billions of dots of the right colour; position does not matter. **Chemistry.**

A quantum computer

A few dots in exactly the right places. **Lithography and voltages.**

A laser

Perfect crystals in a thin layer. **Epitaxy.**

Lecture 2 in one line

Small things are waves; trapped waves have only certain energies; the size of the trap sets those energies; and we now know how to build traps of the size we want.

Part 1: the quantum world. Part 2: wells, wires and dots, and what they do. Part 3: making them. Next, Lecture 3: tunnelling.

Credits

Made for this course

All figures were made for KKKC6353 (scripts in figures-source/). Animation screenshot from faizverse.github.io/nanoelectronic-devices.

Sources

Original L2 handout, Fabrication processes of quantum dots; G. Mishra, Fabrication of quantum dots (slides). Murray, Norris and Bawendi (1993); LaMer and Dinegar (1950); Leonard et al. (1993); Iijima (1991); Arnold et al. (2006); Liu et al. (2020).